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Ans to the question no Q1 (b)

Given, $z = x^2 + 2x + y^2$ — ①

Differentiating ① Partially w.r.t x

$$\frac{\partial z}{\partial x} = 2x + 2 \cdot 1 + 0 \text{ — ②}$$

multiplying ② by x

$$x \frac{\partial z}{\partial x} = 2x^2 + 2x \text{ — ③}$$

Again differentiation ① partially w.r.t y

$$\frac{\partial z}{\partial y} = 0 + 0 + 2y \text{ — ④}$$

multiplying ④ by y

$$y \frac{\partial z}{\partial y} = 2y^2 \text{ — ⑤}$$



Adding equation ③ and ⑤ we get

$$x \frac{\partial z}{\partial x} + y \frac{\partial z}{\partial y} = 2x^2 + 2x + 2y^2$$

$$= 2(x^2 + x + y^2)$$

Ans to the question no Q1(a)

Let $z = x^2 + y^3$

~~$\frac{\partial z}{\partial x} = 2x + 3y^3$~~

$\frac{\partial z}{\partial y} = x^2 + 3y^2$

Ans to the question no 21 (a)

$$z = x^2 + y^3$$

$$\frac{\partial z}{\partial x} = x^2 + y^3 = 2x + y^3$$

$$\frac{\partial z}{\partial y} = x^2 + y^3 = x^2 + 3y^2$$

$$\frac{\partial z}{\partial x} = x^2 + y^3 = x^2 + 3y^2$$

Ans

Ans to the question no. Q2 (a)

P-4

In partial differentiation, one independent variable varies but total differentiation, all the independent variables are varied - total differentiation.

$$\text{let } z = f(x, y) \text{ and } z + \delta z = f(x + \delta x, y + \delta y)$$

$$\delta z = f(x + \delta x, y + \delta y) - f(x, y) \quad \text{--- (1)}$$

now subtracting and adding with the right hand side ab (1) of $f(x + \delta x, y + \delta y)$

$$\delta z = f(x + \delta x, y + \delta y) - f(x + \delta x, y + \delta y) + f(x + \delta x, y + \delta y) - f(x, y + \delta y) + f(x, y + \delta y) - f(x, y)$$

$$= f(x + \delta x, y + \delta y) - f(x + \delta x, y + \delta y) + f(x + \delta x, y + \delta y) - f(x, y + \delta y) + f(x, y + \delta y) - f(x, y)$$

$$+ \frac{f(x + \delta x, y + \delta y) - f(x, y + \delta y)}{\delta x} \delta x + \frac{f(x, y + \delta y) - f(x, y)}{\delta y} \delta y \quad \text{--- (2)}$$

If $\delta x \rightarrow 0$ and $\delta y \rightarrow 0$ then $\delta z \rightarrow 0$

Taking limit in equation (2)

$$\lim_{\delta z \rightarrow 0} \delta z = \lim_{\delta x \rightarrow 0} \frac{f(x+\delta x, y+\delta y) - f(x, y+\delta y)}{\delta x}$$

$$\lim_{\delta x \rightarrow 0} \delta x + \lim_{\delta y \rightarrow 0} \frac{f(x, y+\delta y) - f(x, y)}{\delta y}$$

$$\lim_{\delta y \rightarrow 0} \delta y$$

$$\Rightarrow \delta z = \frac{\partial f}{\partial x} \delta x + \frac{\partial f}{\partial y} \delta y$$

which is called total differential

at z

Ans to the question no Q2 (c)

P-6

$$\text{Given, } z = x^2 + y^2$$

$$\therefore \frac{\partial z}{\partial \theta} = \frac{\partial z}{\partial x} \cdot \frac{dx}{d\theta} + \frac{\partial z}{\partial y} \cdot \frac{dy}{d\theta}$$

$$= 2x \cdot (-r \sin \theta) + 2y \cdot (r \cos \theta)$$

$$= 2r \cos \theta (-r \sin \theta) + 2r \sin \theta (r \cos \theta)$$

$$= -2r^2 \sin \theta \cos \theta + 2r^2 \sin \theta \cos \theta$$

$$= 0$$

$$\text{Again } z = x^2 + y^2 = r^2 \cos^2 \theta + r^2 \sin^2 \theta$$

$$= r^2 (\cos^2 \theta + \sin^2 \theta)$$

$$= r^2 \cdot 1$$

$$= r^2$$

$$\therefore \frac{\partial z}{\partial \theta} = 0 \text{ (verified)}$$

Ans to the question no 3(b)

Given $ax^2 + 2hxy + by^2 = 0$ (1)
 Dividing both side ab (1) by x^2 we get

$$\frac{ax^2}{x^2} + \frac{2hxy}{x^2} + \frac{by^2}{x^2} = 0$$

$$\Rightarrow \frac{a}{b} + \frac{2h}{b} \cdot \frac{y}{x} + \left(\frac{y}{x}\right)^2 = 0$$

$$\Rightarrow \left(\frac{y}{x}\right)^2 + 2 \cdot \frac{h}{b} \cdot \left(\frac{y}{x}\right) + \frac{a}{b} = 0 \quad (2)$$

which is the quadratic equation
 in $\frac{y}{x}$.

Let, m_1, m_2 be the roots ab equation
 (2). then sum ab roots, $m_1 + m_2 = -\frac{2h}{b}$

Product ab roots, $m_1 m_2 = \frac{a}{b}$

Now, the quadratic equation is
equivalent to $\left(\frac{y}{n} - m_1\right) \left(\frac{y}{n} - m_2\right) = 0$

$$\Rightarrow \frac{y}{n} - m_1 = 0$$

$$\frac{y}{n} - m_2 = 0$$

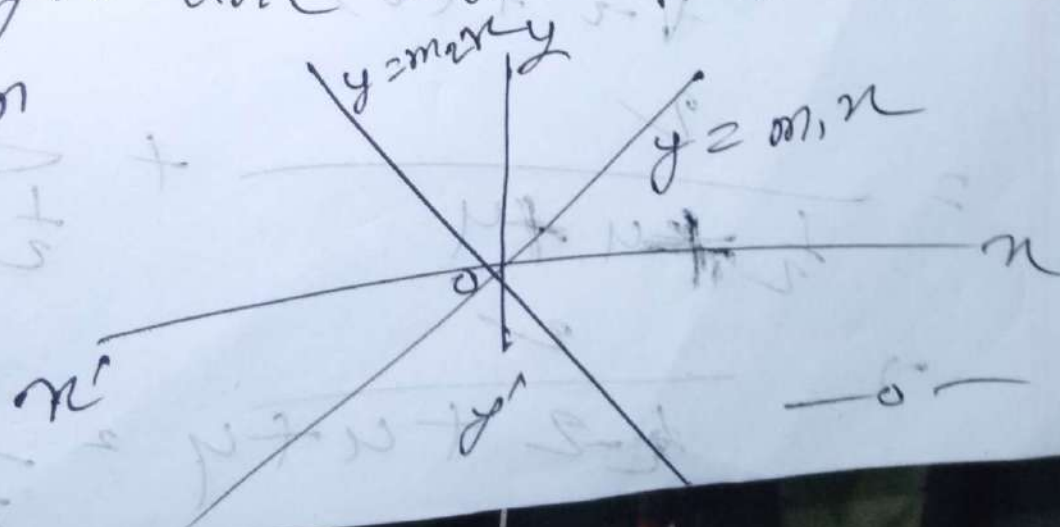
$$\therefore \frac{y}{n} = m_1$$

$$\frac{y}{n} = m_2$$

$$\Rightarrow y = m_1 n$$

$$\Rightarrow y = m_2 n$$

Hence the homogeneous quadratic
equation $ax^2 + 2hxy + by^2 = 0$
represent a pair of
straight lines which pass through
origin



Ans to the question no 24(a)

Is the line is

$$0-2, 0-2, 0-2$$

$$= -2, +2, -2$$

Is the line is

$$\frac{2}{\sqrt{2} + (-2) + (-2)} \quad \frac{2}{\sqrt{-2} + (-2) + (-2)}$$

$$= \frac{2}{\sqrt{-2} + (-2) + (-2)} + \frac{2}{\frac{1}{2} + 4 + 4}$$

$$\rightarrow \frac{2}{-2 + 4 + 4} = \frac{2}{\frac{1}{2} + 8}, \frac{2}{\frac{1}{2} + 8}$$